7-B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING)
COMPONENT WISE DISTRIBUTION

Main Curriculum Components	Sub Components	Approved Credits for B.Tech.	Approved Credits Range	Proposed Credits for B.Tech. by Department	Proposed Credits Range
Institute Core Course	HSSC	5	52-58	**5**	**53**
	HSSEC	6		**6**	
	MC	3		**3**	
	BSC	12-20		**20**	
	ESC	8-20		**8**	
	DSC	4		**4**	
	ESSC	3		**3**	
	TM	4		**4**	
Program Core Course	CCCC	40-48	87-91	**45**	**91**
	AI/ML	2		**2**	
	Engg. Analysis and design (design thinking based project)/Industry Oriented Problem Solving/ Lab based Project/ Practical Problem/ Case study	4		**4**	
	Technical Communication	2		**2**	
	BTP/Entrepreneurship/ Project-based internship/PEC	6-10		**10**	
	PEC	22-26		**22**	
	TEB	6-8		**6**	
	OEC	9-12	9-12	**9-12**	**9-12**
	CORE	2	2	**2**	**2**
	Total	**150-160**		**155-158**	
	MSC/DHC	**18/20**		**18/20**	
	Grand Total			**173/178**	

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

Program Code : 116 B.Tech. (Electronics & Communication Engineering)
Department : ECE Electronics & Communication Engineering

Teaching Scheme

Year	Credits in Autumn Semester	Credits in Spring Semester	Credits (Year – wise)
1	23	21	44
2	23/24	23/24	46/48
3	21/22	18/22	39/44
4	12/20	14/22	26/42
Grand Total			**155/178**
Total with MSC/DHC	**With addition 18-20 credits**		**173/198**

	Components	Maximum Units	Minimum Units	Comments
Non-Credit Elements (NCE)	**Discipline (DIS)**	**16**	**8**	To be evaluated by DoSW
	NCC/NSS/NSO	**8**	**4**	To be evaluated by DoSW
	Internship (INT)	**24**	**8**	1-week internship= 1 unit (to be coordinated by the deptt. /Centres/School)
	Participation in professional development programs by Industry experts/ field experts (PPD-1 & PPD-2)	**8**	**4**	To be coordinated by the departments/Centres/school (2nd& 3rd Years)
Minimum non-credit units to be earned: 24				

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Program Code : 116 - B.Tech. (Electronics and Communication Engineering)
Department : Department of Electronics and Communication Engineering
Year : I

Teaching Scheme					Contact Hours/Week			Exam Duration (Hrs.)		Relative Weights(%)				
S. No.	**Subject Code**	**Course Title**	**Subject Area**	**Credits**	**L**	**T**	**P**	**Theory**	**Practical**	**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
(Autumn)														
1	HSI-101	Soft Skills	HSSC	3	2	0	2	2	0	10-25	25	15-25	30-40	-
2	MAI-101	Mathematics-I	BSC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	PHI-101	Physics-I	BSC	4	3	1	2/2	3	0	15-30	20	15-25	30-40	-
4	ECC-101	Introduction to Programming	PCC	4	3	0	2/2	3	0	15-30	20	15-25	30-40	-
5	TMI-101	Tinkering and Mentoring	TMI	4 T-2	-	-	-	-	-	70	30	-	-	-
				4 M-2	2	0	0	2	-	50	-	-	50	-
6	CSE-101	Data Structures and Algorithms	ESC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
		Total		**23**										
(Spring)														
1	HSI-102	Indian Knowledge System	HSSC	2	2	0	0	2	0	20-35	-	20-30	40-50	-
2	MAI-102	Mathematics-II	BSC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	ESS-102	Environmental Science and Sustainability	ESSC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
4	ECC-102	Semiconductor Devices and Applications	PCC	4	3	0	2/2	3	0	15-30	20	15-25	30-40	-
5	ECC-104	Digital Logic and Systems	PCC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
6	ECC-106	Signals and Systems	PCC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
		Total		**21**										

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Program Code : 116 - B.Tech.(Electronics and Communication Engineering)
Department : Department of Electronics and Communication Engineering
Year : II

Teaching Scheme					Contact Hours/Week			Exam Duration (Hrs.)		Relative Weights (%)				
S. No.	**Subject Code**	**Course Title**	**Subject Area**	**Credits**	**L**	**T**	**P**	**Theory**	**Practical**	**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
(Autumn)														
1	MSI-101	Fundamentals of Management	MC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
2	MAB-103	Numerical Methods	BSC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	ECC-201	Fundamentals of Communication System	PCC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
4	ECC-203	Electromagnetics and Radiating Systems	PCC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
5	ECC-205	Network Theory	PCC	2	2	0	0	2	0	20-35	-	20-30	40-50	-
6	HSSEC-I	HSS Elective Course-I	HSSEC	3										
7	OEC-I	Open Elective Course-I	OEC	3/4										
		Total		**23/24**										
(Spring)														
1	DAI-101	Data Science	DSC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
2	PHB-102	Quantum and Statistical Mechanics	BSC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	EEE-101	Control System Engineering	ESC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
4	ECT-I	Talent Enhancement Course-I	TEB	2	0	0	4	-	-	-	50	-	-	50
5	ECC-202	Digital Communications	PCC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
6	ECC-204	Analog Circuits	PCC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
7	OEC-II	Open Elective Course-II	OEC	3/4										
		Total		**23/24**										

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Program Code : 116 - B.Tech.(Electronics and Communication Engineering)
Department : Department of Electronics and Communication Engineering
Year : III

Teaching Scheme					Contact Hours/Week			Exam Duration (Hrs.)		Relative Weights (%)				
S. No.	**Subject Code**	**Course Title**	**Subject Area**	**Credits**	**L**	**T**	**P**	**Theory**	**Practical**	**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
(Autumn)														
1	ECC-351	Fundamentals of AI/ML	PCC	2	2	0	0	2	0	20-35	-	20-30	40-50	-
2	ECC-301	Digital Signal Processing	PCC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
3	ECC-303	Microwave Engineering	PCC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
4	ECC-305	Digital Electronics and Linear IC Lab	PCC	2	0	0	3	-	-	-	50	-	-	50
5	ECC-399	Community Outreach	CORE	2	0	0	3	-	-	100				
6	ECT-II	Talent Enhancement Course-II	TEB	2	0	0	4	-	-	-	50	-	-	50
7	HSSEC-II	HSS Elective Course-II	HSSEC	3										
8	OEC-III	Open Elective Course-III	OEC	3/4										
		Total		**21/22**										
(Spring)														
1	ECL-I	Program Elective Course - I	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
2	ECL-II	Program Elective Course - II	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
3	ECC-300	Lab Based Project/Industry oriented problem solving	PCC	4	0	1	6	-	-	-	50	-	-	50
4	ECC-302	Microwave Laboratory	PCC	2	0	0	4	-	-	-	50	-	-	50
5	ECC-304	Communication Systems Lab	PCC	2	0	0	4	-	-	-	50	-	-	50
6	ECC-306	Technical Communication	PCC	2	0	2	0				100			
7	ECT-III	Talent Enhancement Course-III	TEB	2	0	0	4	-	-	-	50	-	-	50
8	MSC/DHC - I	Minor Specialization Course - I / Departmental Honours Course - I	MSC/ DHC	3/4										
		Total		**18/21-22**										

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Program Code : 116 - B.Tech.(Electronics and Communication Engineering)
Department : Department of Electronics and Communication Engineering
Year : IV

Teaching Scheme					Contact Hours/Week			Exam Duration (Hrs.)		Relative Weights (%)				
S. No.	**Subject Code**	**Course Title**	**Subject Area**	**Credits**	**L**	**T**	**P**	**Theory**	**Practical**	**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
(Autumn)														
1	ECP-400A/ ECL	BTP/Project/Internship/Entrepreneurship/PEC *	PCC/PEC*	4						100				
2	ECL-III	Program Elective Course -III	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	ECL-IV	Program Elective Course -IV	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
4	MSC/DHC-II	Minor Specialization Course -II /Departmental Honours Course - II	MSC/ DHC	3/4										
5	MSC/DHC-III	Minor Specialization Course -III /Departmental Honours Course - III	MSC/ DHC	3/4										
		Total		**12/18-20**										
(Spring)														
1	ECP-400B/ECL	BTP/Project/Internship/Entrepreneurship/PEC *	PCC/PEC*	6						100				
2	ECL-V	Program Elective Course -V	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
3	ECL-VI	Program Elective Course -VI	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
4	MSC/DHC-IV	Minor Specialization Course -IV /Departmental Honours Course – IV	MSC/ DHC	3/4										
5	MSC/DHC-V	Minor Specialization Course - IV/Departmental Honours Course - IV	MSC/ DHC	3/4										
		Total		**14/20-22**										

List of Program Elective Courses

Teaching Scheme					Contact Hours/Week			Exam. Duration		Relative Weight (%)				
S. No.	**Sub Code**	**Course Title**	**Subject Area**	**Credits**	**L**	**T**	**P**	**Theory**	**Practical**	**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
1.	ECL-343	Fundamentals of Microelectronics	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
2.	ECL-360	Introduction to Information and Communication Theory	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
3.	ECL-347	Introduction to Microwave Semiconductor Device Modelling Techniques	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
4.	ECL-316	Digital Image processing	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
5.	ECL-357	Electronics Subsystems	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
6.	ECL-358	Machine Learning in Semiconductor Manufacturing	PEC	3	3	0	0	3	0	20-35	-	20-30	40-50	-
7.	ECL-514	Detection and Estimation Theory	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
8.	ECL-614	Adaptive Signal Processing Techniques	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
9.	ECL-618	Wireless Technologies: 5G and Beyond	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
10.	ECL-619	Introduction to Compressed Sensing	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
11.	ECL-620	Advanced Wireless Communication	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
12.	ECL-577	VLSI Technology	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
13.	ECL-584	Mixed Signal Circuits	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
14.	ECL-587	Nanoscale Devices	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
15.	ECL-591	VLSI Physical Design	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
16.	ECL-561	Compact Modelling of Semiconductor Devices	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
17.	ECL-526	Statistical Machine Learning for Variation-Aware Electronic Device and Circuit Simulation	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
18.	ECL-635	Magnetic Random Access Memory	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
19.	ECL-550	Radar Signal Processing	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
20.	ECL-631	RF Receiver Design	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
21.	ECL-515	Information and Coding Theory	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
22.	ECL-519	Wireless Communication Systems	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-

23.	ECL-525	Hardware Architecture for deep-learning	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
24.	ECL-573	Digital VLSI Circuit Design	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
25.	ECL-579	Foundations of Semiconductor Device Physics	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
26.	ECL-581	Analog VLSI Circuit Design	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
27.	ECL-542	MicrowaveIntegratedCircuits	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-
28.	ECL-544	AdvancedRadarEngineering	PEC	4	3	1	0	3	0	20-35	-	20-30	40-50	-

List of Talent Enhancement Course

Teaching Scheme					Contact Hours/Week			Exam Duration		Relative Weight (%)				
S. No.	Course Code	Course Title	Area	Cr.	L	T	P	Th.	Pr.	CWS	PRS	MTE	ETE	PRE
RFM-1 (TEB-A)														
1	ECT-101	RF Simulation Techniques	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-102	Design of Passive Microwave Components	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-103	Microwave Measurement Techniques	TEB	2	0	0	4	-	-	-	50	-	-	50
CNSP-1 (TEB-B)														
1	ECT-104	Communication Networks	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-105	Optical Communication Technologies	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-106	Software Defined Radios	TEB	2	0	0	4	-	-	-	50	-	-	50
CNSP-2 (TEB-C)														
1	ECT-107	Introduction to Bio-Signal and Image Analysis	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-108	Principles and Techniques of FMCW Radar Systems	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-109	Advanced DSP	TEB	2	0	0	4	-	-	-	50	-	-	50
MVLSI-1 (TEB-D)														
1	ECT-110	Digital System Design Using High Level Language	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-111	VLSI Circuit Design	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-112	Physical Design	TEB	2	0	0	4	-	-	-	50	-	-	50
MVLSI-2 (TEB-E)														
1	ECT-113	Analog Circuit Design	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-114	Mixed Signal System Design	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-115	Measurement Techniques	TEB	2	0	0	4	-	-	-	50	-	-	50

MVLSI-3 (TEB-F)														
1	ECT-116	Fabrication Techniques	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-117	Semiconductor Characterization	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-118	Device Modeling	TEB	2	0	0	4	-	-	-	50	-	-	50
MVLSI-4 (TEB-G)														
1	ECT-119	Microprocessor and Computer Architecture	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-120	VLSI DSP	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-121	Embedded Systems	TEB	2	0	0	4	-	-	-	50	-	-	50
MVLSI-5 (TEB-H)														
1	ECT-122	Emerging Materials and Devices	TEB	2	0	0	4	-	-	-	50	-	-	50
2	ECT-123	Memories	TEB	2	0	0	4	-	-	-	50	-	-	50
3	ECT-124	Emerging Computing Techniques	TEB	2	0	0	4	-	-	-	50	-	-	50

Minor Specializations Courses (20 credits)

S.No.	Code	Course title	Semester	Credits
1	ECC-102	Semiconductor Devices and Applications	Spring	4
2	ECC-104	Digital Logic and Systems	Spring	4
3	ECC-106	Signals and Systems	Autumn	4
4	ECC-201	Fundamentals of Communication System	Autumn	4
5	ECC-203	Electromagnetics and Radiating Systems	Spring	4
6	ECC-202	Digital Communications	Spring	3
7	ECC-204	Analog Circuits	Autumn	3
8	ECC-303	Microwave Engineering	Autumn	4
9	ECC-301	Digital Signal Processing	Autumn	3

Departmental Honours Courses (20 credits)

S.No.	Code	Course title	Semester	Credits
1	ECL-614	Adaptive Signal Processing Techniques	Spring	4
2	ECL-618	Wireless Technologies: 5G and Beyond	Spring	4
3	ECL-620	Advanced Wireless Communication	Autumn	4
4	ECL-577	VLSI Technology	Autumn	4
5	ECL-584	Mixed Signal Circuits	Spring	4
6	ECL-573	Digital VLSI Circuit Design	Spring	4
7	ECL-542	Microwave Integrated Circuits	Autumn	4
8	ECL-544	Advanced Radar Engineering	Autumn	4
9	ECL-631	RF Receiver Design	Autumn	4